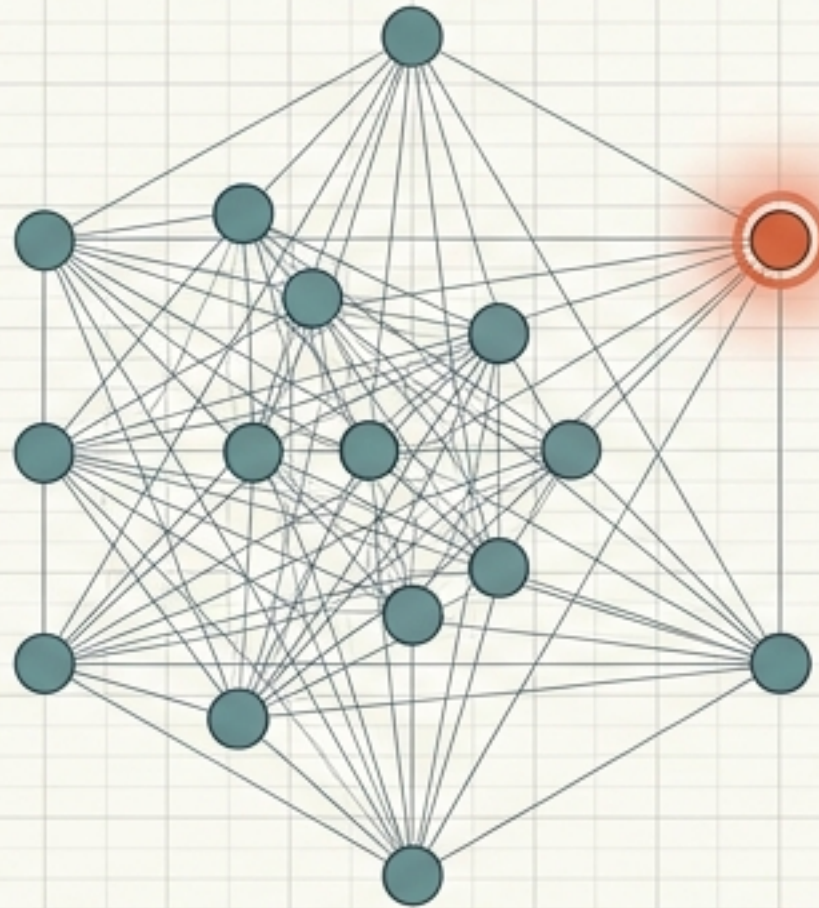


The Architecture of Inquiry



A visual primer on biostatistics, study design,
and the pursuit of statistical truth.

Variable: A recorded characteristic.

student	gender	sleep	bedtime	countries	dread
1	female	5	12-2	0	3
2	female	7	10-12	0	2
3	male	5.5	12-2	1	4
4	female	7	12-2		2
5	female	3	2-4	0	3
6	female	3	2-4	1	3
7	male	8	10-12	0	4
8	female	7	10-12	0	3
9	female	8	10-12	0	3
10	female	7	12-2	1	4

Observation: A single entity's complete dataset.

Every statistical inquiry begins here. Before we can infer truths about a population, we must structure our observations into a standardized matrix of variables.

Variables

Numerical



Continuous

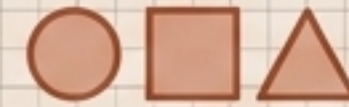
(Measured characteristics)
e.g., weight, time, height.



Discrete

(Counted characteristics)
e.g., age, # of children.

Categorical



Nominal

(Defined categories
without order)
e.g., gender, eye color,
political party.



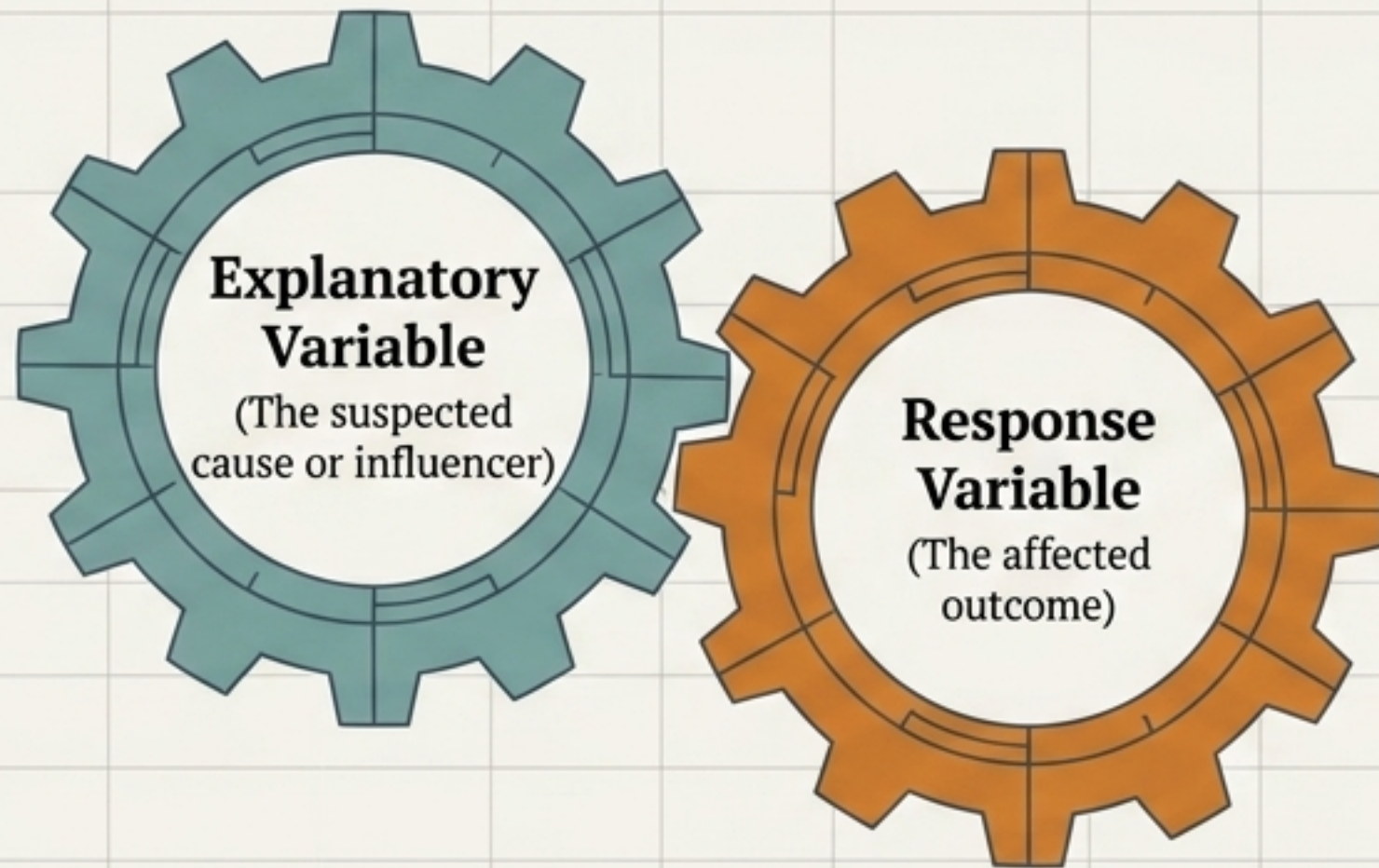
Ordinal

(Ordered categories)
e.g., ratings (low,
medium, high; 1-5).

The Area Code Trap

Q: Is a telephone area code numerical?
A: No. It is Categorical (Nominal). Even though it uses numbers, the numbers don't represent magnitude.

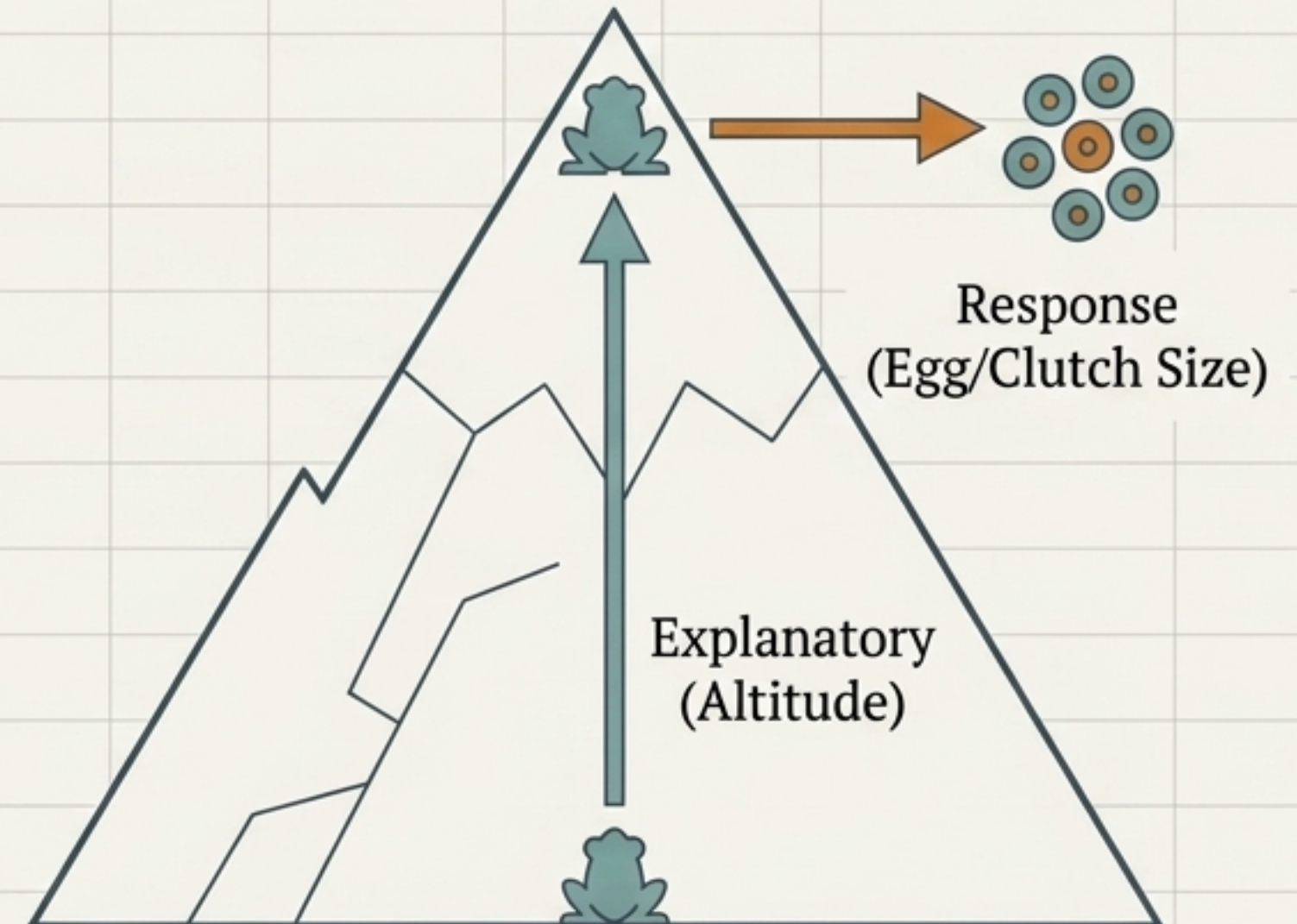
The Concept



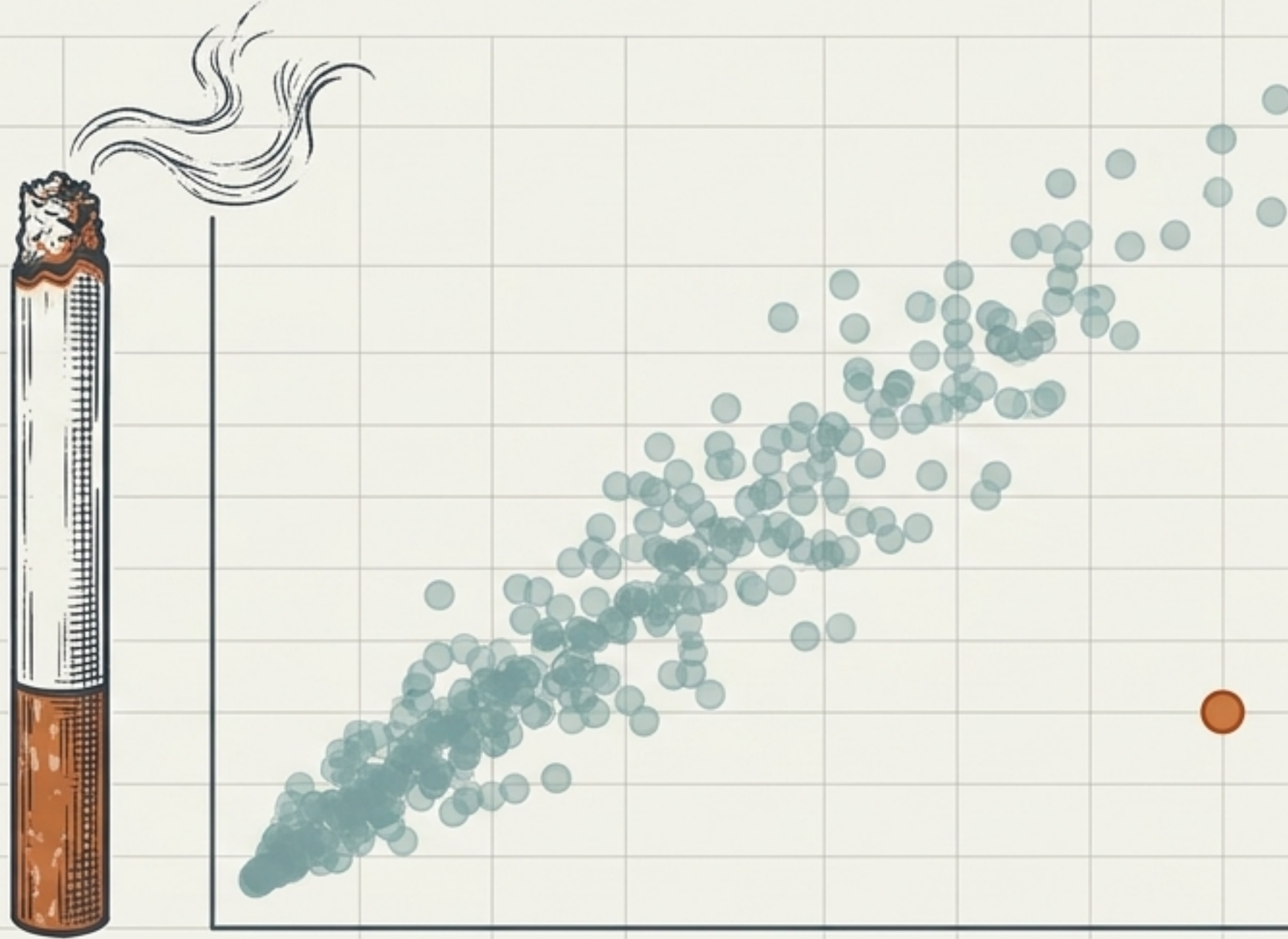
Caveat: Labeling these does **NOT** guarantee the relationship is truly **causal**.

Case Study: Frog Maternal Investment

Researchers measured egg clutches at 11 sites at differing altitudes.



The Anecdote Illusion



“In the 1930s, anti-smoking research faced resistance from arguments like: ‘My uncle smokes three packs a day and he’s in perfectly good health.’”

1. **Sample Size of One:** The data represents an isolated, single case.
2. **Unrepresentative:** Unclear if this isolated individual represents the broader population.
3. **Human Variability:** Complex biological behaviors require immense samples to reveal true trends through the noise.

Takeaway: Anecdotal evidence is a haphazard collection of single data points. It may be true, but it is never a foundation for generalized truth.

Study Design: Observational Studies vs. Experiments

Observational Studies

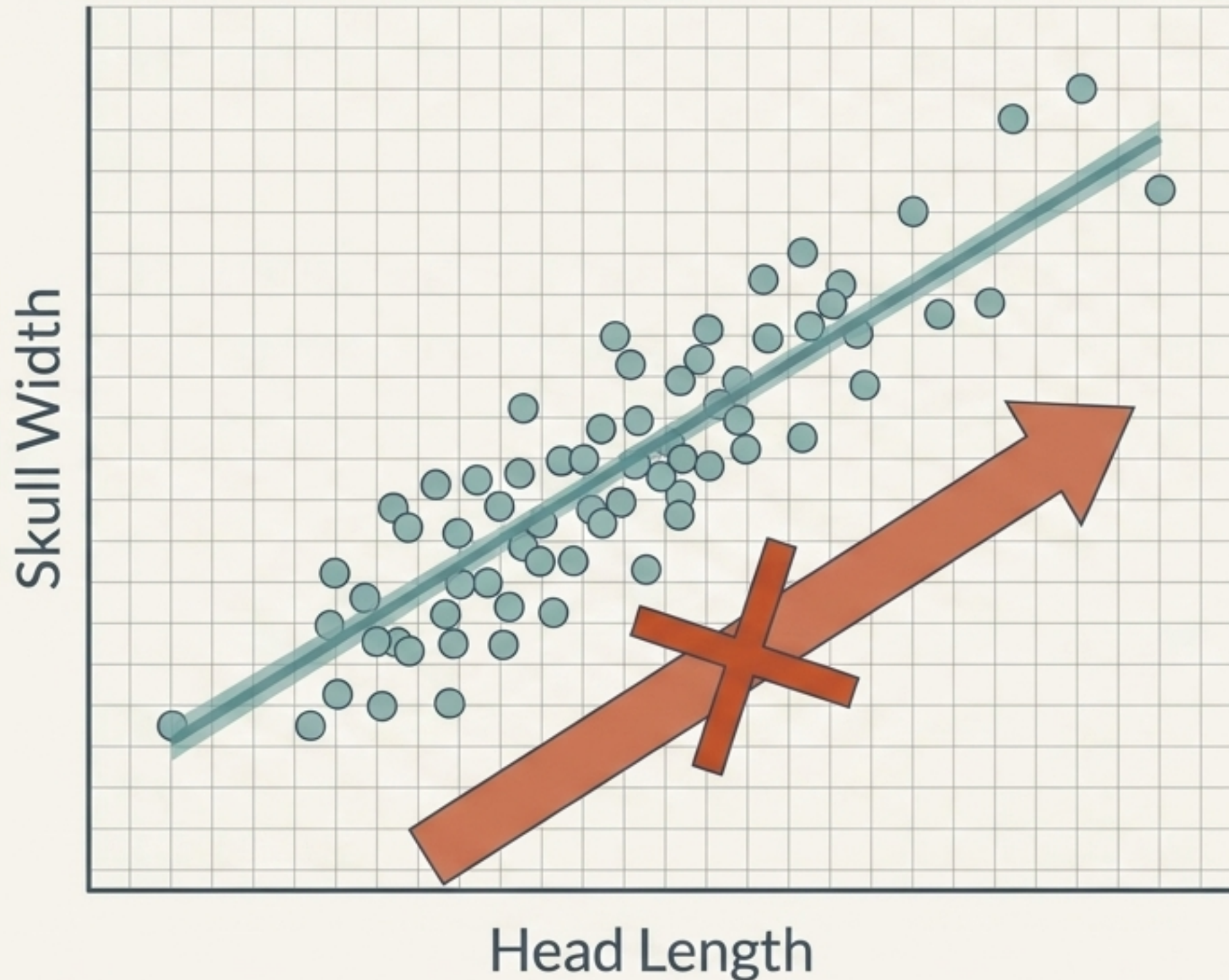
- **Mechanism:**
Researchers collect data without interfering with how it arises.
- **Output:**
Establishes **ASSOCIATION** (not causation).
- **Example:**
Asking a random sample about bedtime habits to see if tea drinkers naturally sleep earlier.

Experiments

- **Mechanism:**
Researchers randomly assign subjects to different treatments.
- **Output:**
Establishes **CAUSATION**.
- **Example:**
Randomly forcing one group to block social media and another to use it normally to measure resulting happiness.

Causation can ONLY be inferred from a randomized experiment.

Association $\neq$ Causation



The Observation:

The data shows a clear positive association. As head length increases, skull width tends to increase.

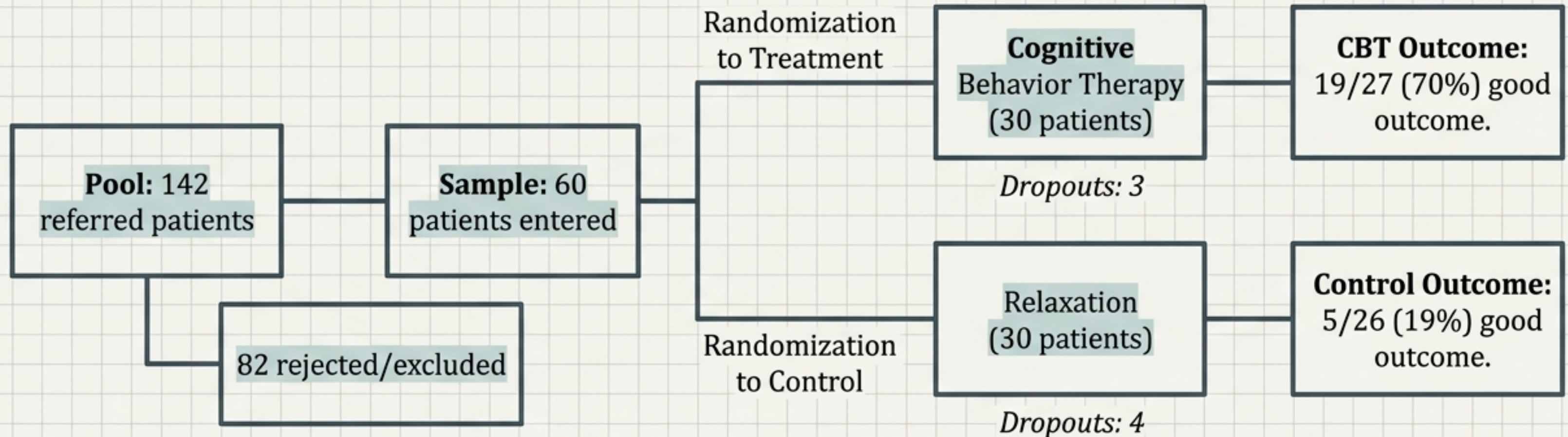
The Trap:

The Trap: It is tempting to look at the trendline and conclude “A longer head causes the skull to be wider.”

The Reality:

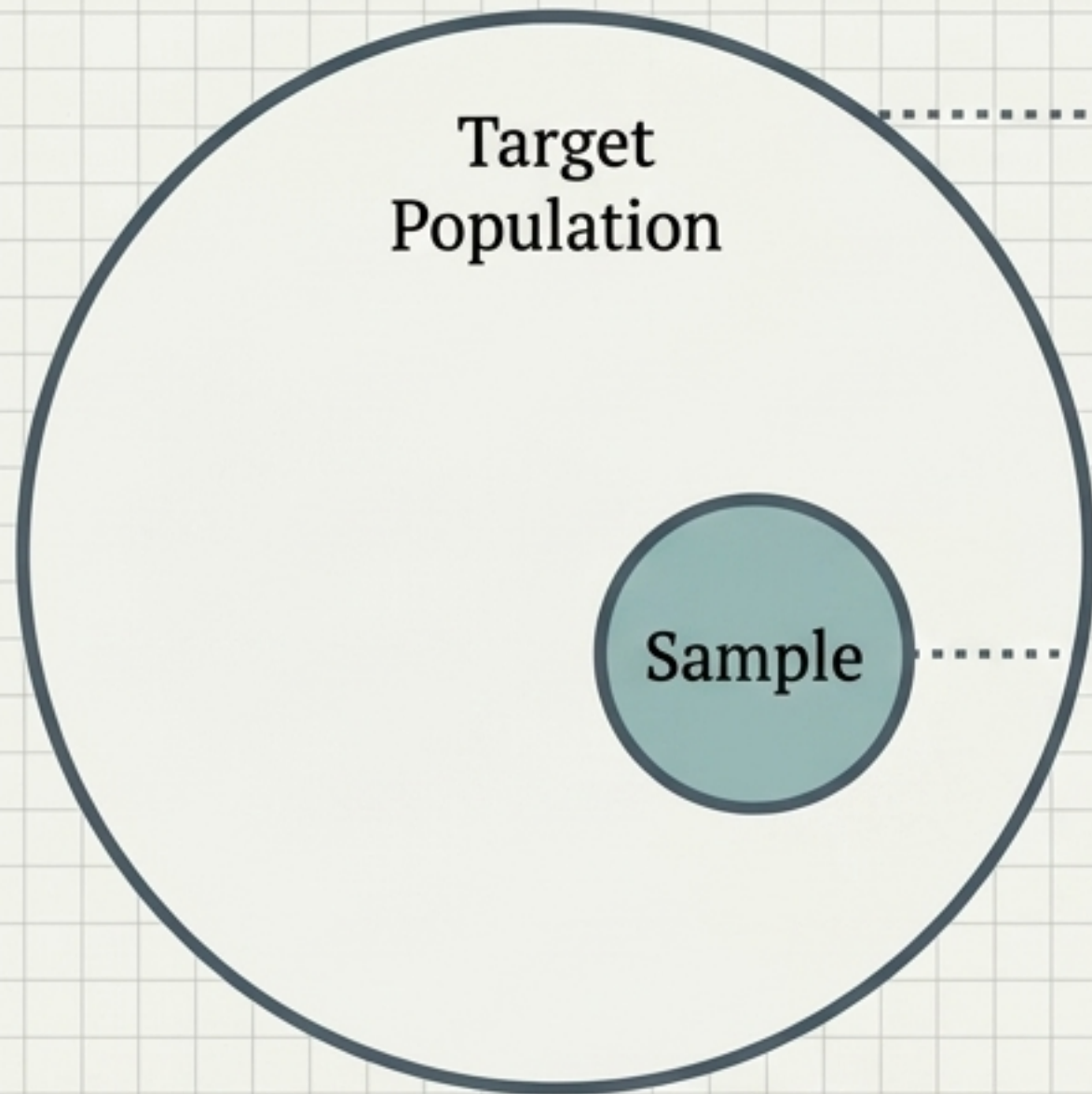
The Reality: They are merely associated (dependent) variables. Because this is an observational study of natural biology, we cannot infer a causal link.

Experimental Pipeline: Randomized Clinical Trial Flowchart



The difference in outcome is 51%. Is this real or just “natural variation” (like flipping 50 heads in 100 coin tosses)? Because the statistical gap is so massive, we can confidently infer the difference is caused by the treatment.

Target Population vs. Sample & The Census Problem



Target Population:

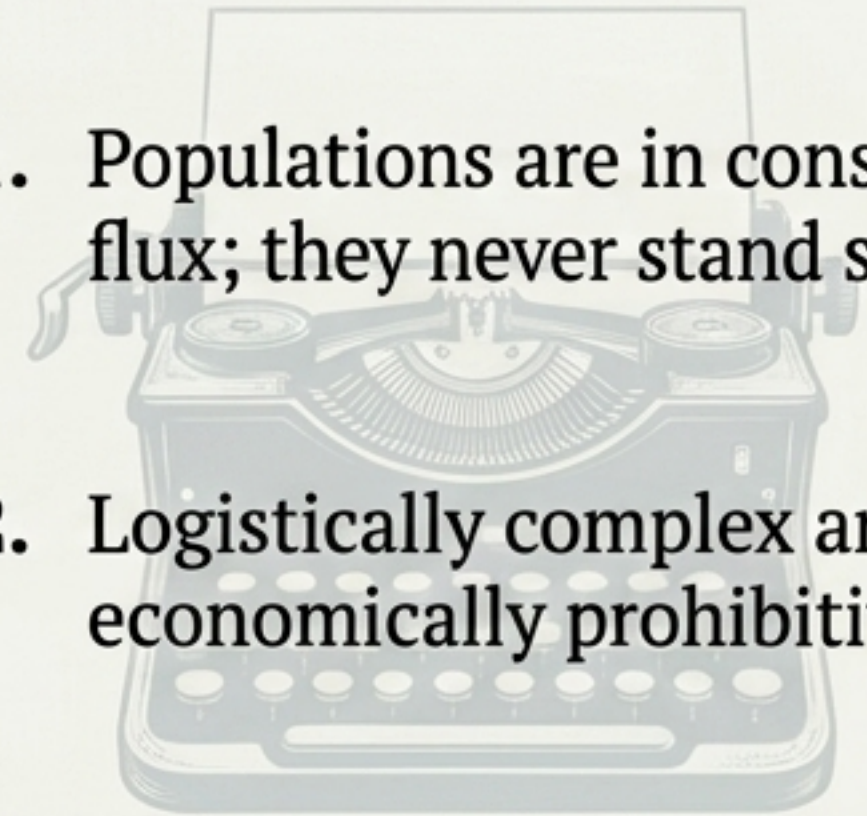
All entities referring to the research question (e.g., “All people” when asking if milk makes bones stronger).

Sample:

A small fraction of the population actually measured (e.g., “A specific group of adults drinking milk”).

The Census Problem (Why not sample everyone?)

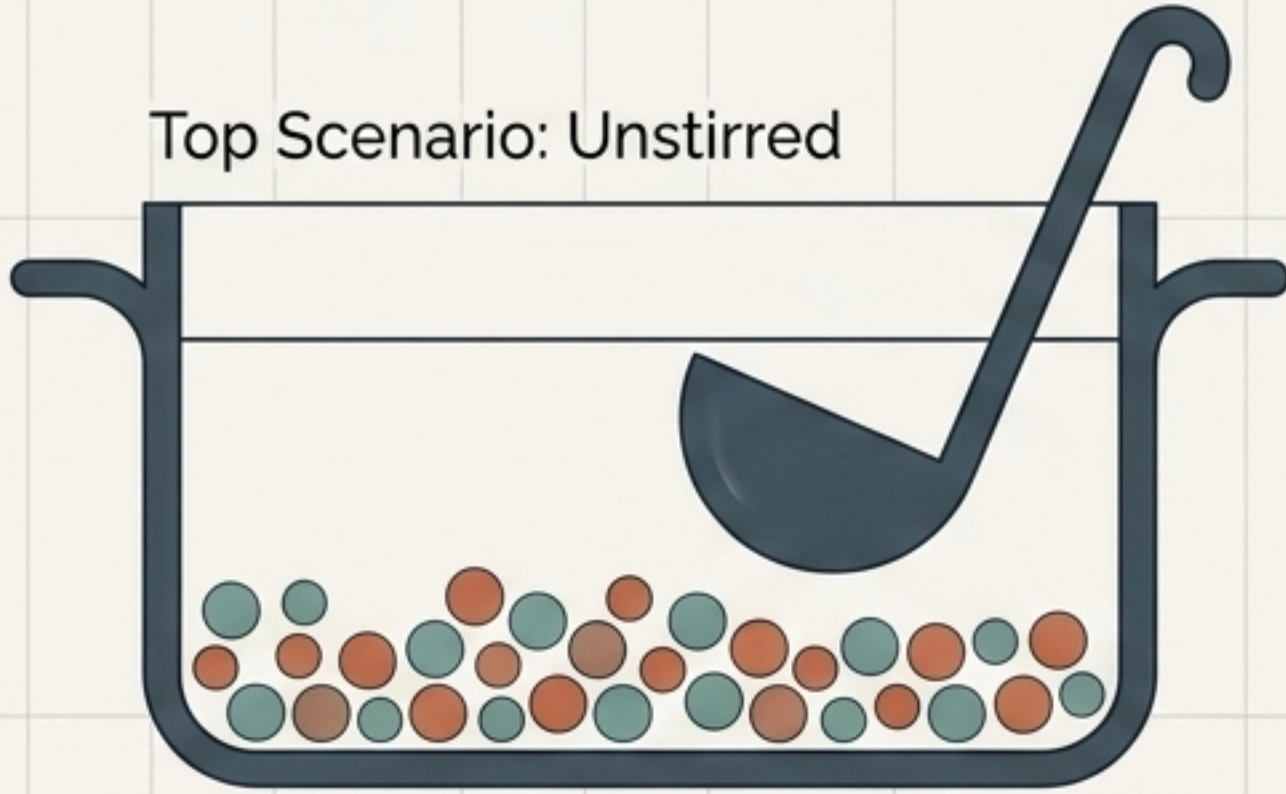
1. Populations are in constant flux; they never stand still.
2. Logistically complex and economically prohibitive.



Generalizability relies entirely on the sample being randomly selected from the target population.

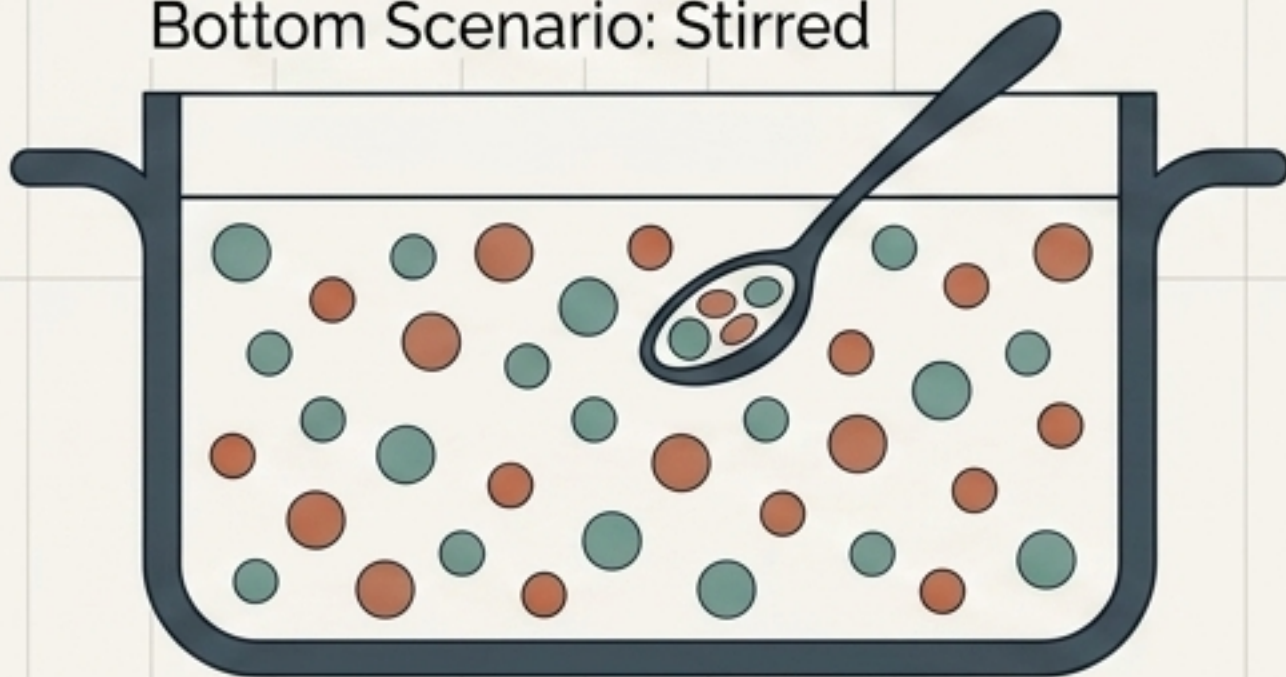
The Soup Analogy (Bias vs. Size)

Top Scenario: Unstirred



Large Sample + Biased Collection (Unstirred)
= An inaccurate taste of the recipe.

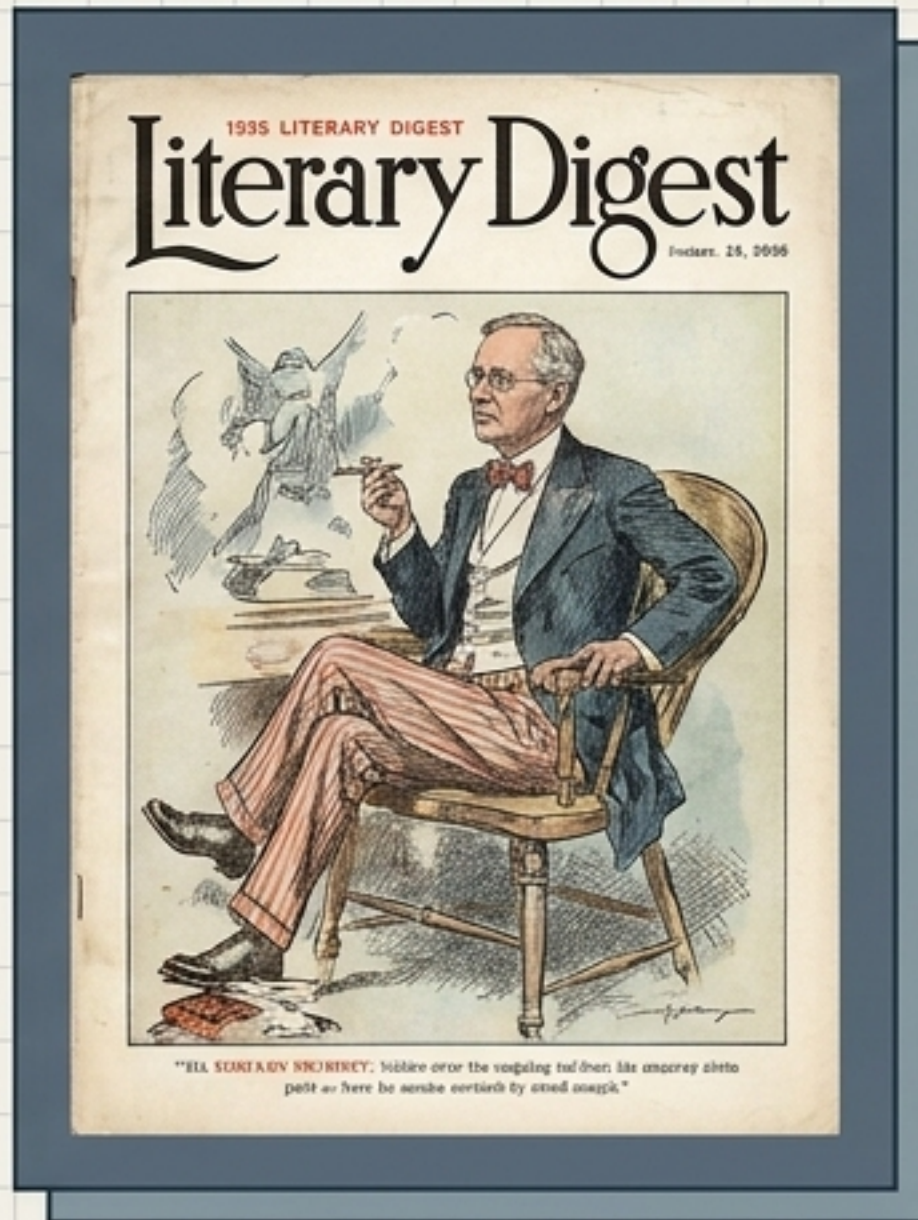
Bottom Scenario: Stirred



Small Sample + Random Collection (Stirred)
= A perfect representation of the whole.

A massive sample size cannot cure sampling bias. If the methodology is flawed, a larger sample merely gives you a more precise measurement of the wrong answer.

Autopsy of a Failure: The 1936 Literary Digest Poll



The Setup: The magazine polled a massive 10 million Americans, receiving 2.4 million responses.

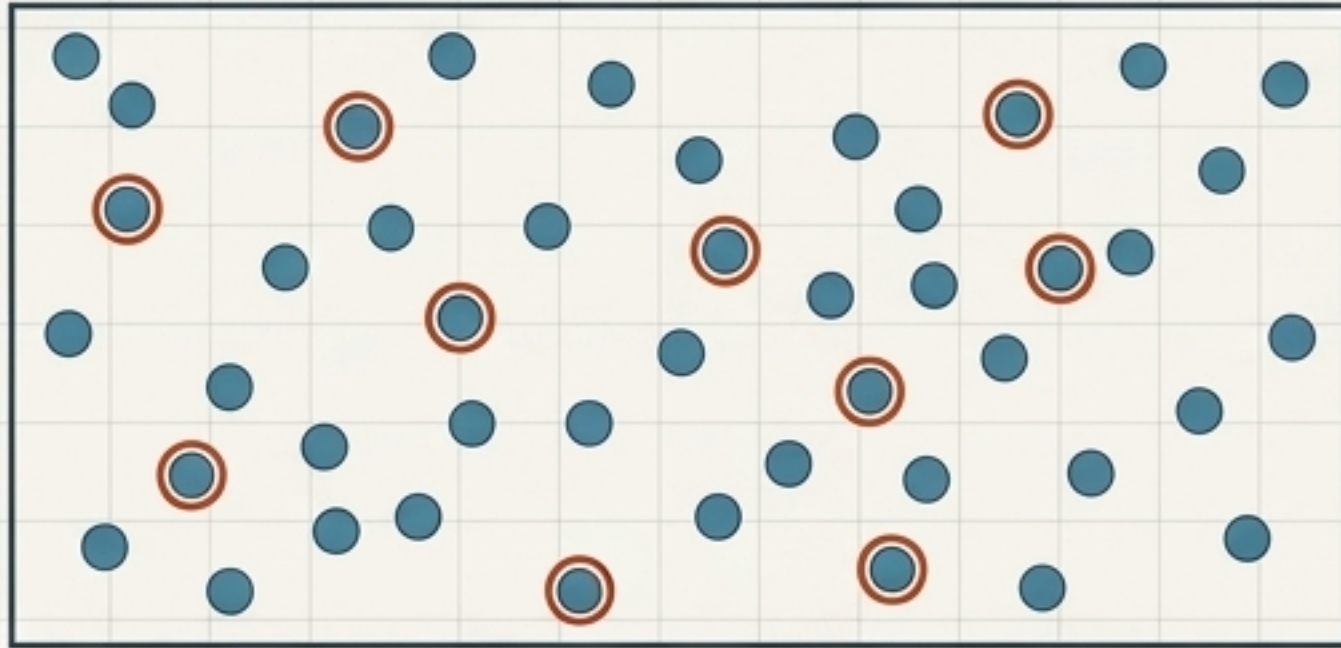
The Prediction: A landslide victory for Republican Alf Landon over FDR.

The Reality: FDR won in a historic landslide (62%). The magazine was completely discredited and folded.

The Autopsy (Why it failed)

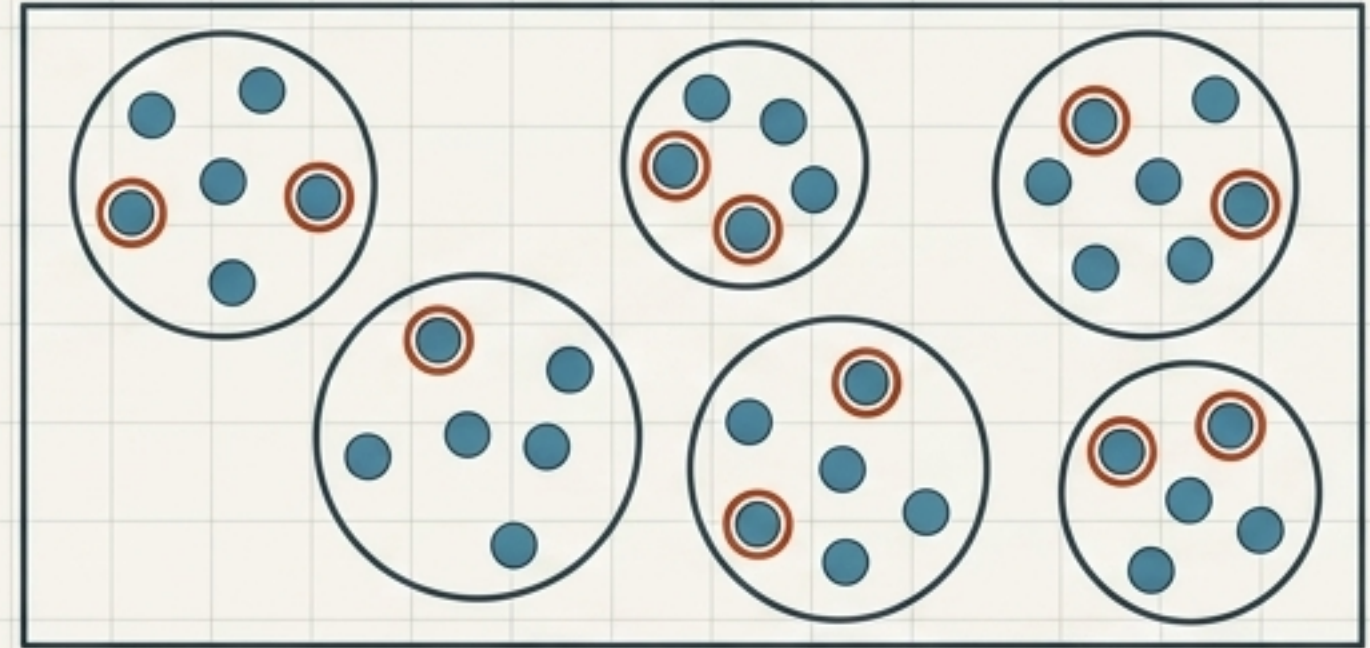
- **The Bias:** They surveyed their own readers, registered auto owners, and telephone users.
- **The Context:** In the depths of the Great Depression, this methodology effectively restricted the sample to the **wealthy**, who heavily favored **Republicans**.
- **The Verdict:** The soup was completely **unstirred**. The massive 2.4 million sample size was entirely irrelevant because the initial collection method was **structurally biased**.

Simple Random



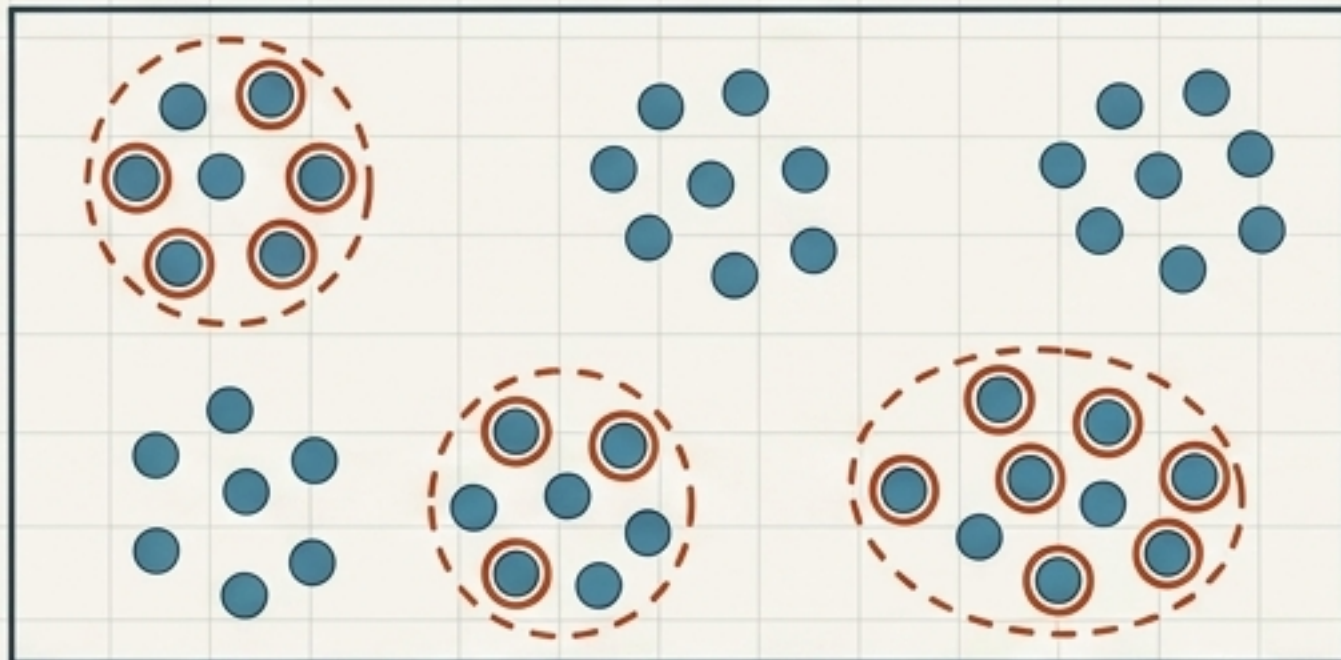
No implied connection between selected points.

Stratified



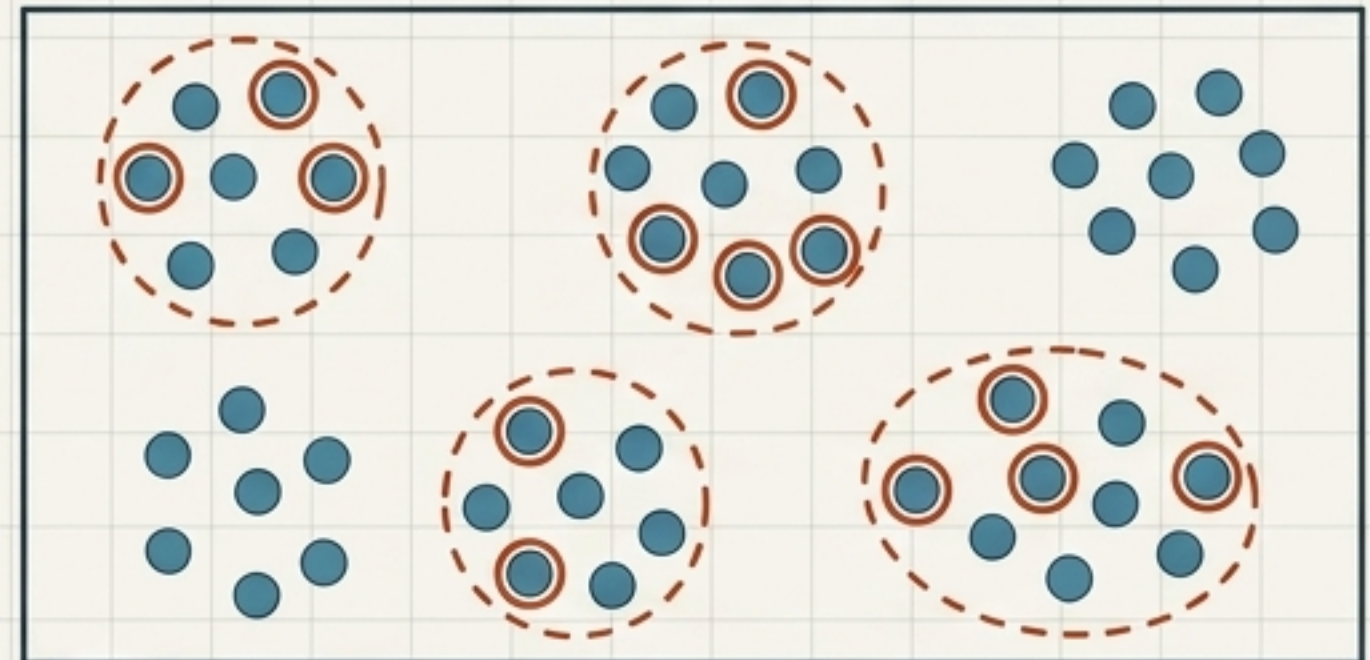
Divide into similar groups, sample some from all.

Cluster



Divide into mixed groups, sample all from some.

Multistage



Sample clusters, then sample within the clusters.

Stratified vs. Cluster Sampling

	Stratified Sampling	Cluster Sampling
The Grouping Logic	Groups (Strata) are HOMOGENOUS (similar observations inside each group).	Groups (Clusters) are HETEROGENEOUS (each group is a mini-representation of the whole).
The Selection Method	We select SOME individuals from ALL groups.	We select ALL individuals from SOME groups.
Primary Benefit	Ensures minority populations and distinct sub-groups are proportionally represented.	Highly economical and logistically easier when dealing with large, spread-out populations.

